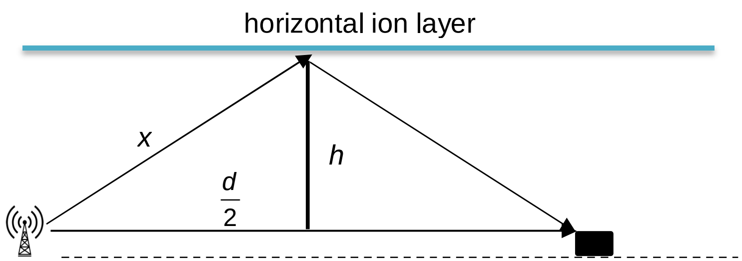
	$\sqrt{\langle c^2 \rangle_{He}} = 2.24\ v$	[1] answer
4(a)	Stationary waves are formed by the <u>superposition of two progressive waves of equal amplitude, frequency and speed</u> but <u>travelling in opposite directions</u>.	[1] equal amplitude, frequency, speed [1] opposite directions

4(b)(i)	$v = f\lambda \Rightarrow \lambda = \frac{v}{f} = \frac{3.0\times10^{8}}{9.0\times10^{7}} = 3.33$ m Distance between 2 nodes = $= \frac{\lambda}{2} = \frac{3.33}{2} = 1.67$ m	[1] correct λ [1] answer
4(b)(ii)	Distance travelled by the receiver in 1 s = 6.0×10^{2} m. Number of nodes detected in 1 s = $\frac{6.0\times10^{2}}{1.67} = 360$ nodes s^{-1}.	[1] [1]
4(b)(iii)	 As there is a phase change of π radians at reflection, to get destructive interference, the path difference is $n\lambda$. Using Pythagoras theorem, $x = \sqrt{h^2 + \left(\frac{d}{2}\right)^2}$ Path difference between the direct signal and reflected signal $= 2x - d$ $= 2\sqrt{h^2 + \left(\frac{d}{2}\right)^2} - d$ For minimum height, n = 1. $= 2\sqrt{h^2 + \left(\frac{d}{2}\right)^2} - d = \lambda$ $\Rightarrow 2\sqrt{h^2 + \left(\frac{10000}{2}\right)^2} - 10000 = 3.33$ $h = 129$ m $= 130$ m	[1] path difference = $n\lambda$. [1] correct x [1] correct path difference [1] correct h
5(a)	E.m.f. is the work done/energy required to move a unit charge	[1] correct